

DANIEL LIAN

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🐙 github.com/GlutenFreeGrapes

Highly motivated graduate student in Computer Science with passion for software engineering, artificial intelligence, machine learning, cybersecurity, and data science. Outside of computer science, I enjoy running and volleyball.

Education

University of Michigan

Master of Science in Engineering in Computer Science & Engineering

Aug. 2026 – May 2027

Ann Arbor, Michigan

University of Michigan

Bachelor of Science in Engineering in Computer Science - Summa Cum Laude (3.76/4.00)

Aug. 2023 – May 2026

Ann Arbor, Michigan

Relevant Coursework

- | | | | |
|-----------------------|---------------------|-----------------------------------|-----------------------------|
| • Conversational AI | • Algorithms | • Cryptography | • Operating Systems |
| • Ethics and AI | • Software Design | • Foundations of Computer Science | • Computer Organization |
| • Computer Vision | • Problem Solving | • Discrete Math | • Computational Linguistics |
| • Data Structures and | • Data Science | | |
| | • Computer Security | | |

Experience

United States Patents and Trademark Office

AI Patents Intern

June 2026 – August 2026

McLean, VA

- Examined artificial intelligence-related patents as part of Technology Center 2100, evaluating novel machine learning architecture and deep learning methods
- Developed application to help speed up search process

Fairfax Collegiate

Head Instructor

June 2023 – August 2025

McLean, VA

- Led 15 STEM classes over 15 weeks, teaching over 100 students from 3rd grade through 10th grade robotics, Scratch programming, advanced math concepts (including middle school math and algebra), and game design.
- Honed communication skills by mentoring junior staff, created friendlier team-based learning environment to adapt to students' diverse learning styles.

Projects

Compas Course Recommender | *TypeScript, HTML, PostgreSQL, Python, Selenium*

January 2026 - May 2026

- Designed and developed system using Gemini backend LLM to provide University of Michigan course recommendations and reasoning based on user's degree program, course history, and preferences with iterative feedback mechanism.
- Used PostgreSQL backend to store course data from over 14,000 courses with metrics for interest and workload.

Cities Chain Discord Bot | *Python, Pandas, MariaDB, SQL*

July 2023 - Present

- Designed and implemented a scalable Discord game supporting thousands of players across 1000+ Discord servers using efficient SQL backend data management in MariaDB for seamless gameplay.
- Applied Agile software maintenance and iterative feedback loops to ensure robust gameplay experience.

Subway Challenge | *Python, Pandas*

December 2022 - May 2024

- Collected and analyzed GTFS data for metropolitan rail systems across 6 cities (Kochi, Los Angeles, Atlanta, San Francisco, Washington, New York), created graph structure using schedule data to model regular routes.
- Implemented heuristics-based breadth-first search algorithm to find fastest complete traversals of all metro systems.

Election Prediction Algorithm | *Jupyter Notebook, TensorFlow, Keras, Pandas, NumPy*

August 2022 - May 2023

- Implemented Python Tensorflow model in Jupyter Notebook to predict 85% of 2022 United States congressional winners using weighted average of 538 polls, demonstrating robust predictive analytics capabilities.
- Presented findings as part of senior research project to audience of 40 at tjSTAR technology research symposium.

Technical Skills

Languages: Python (Pytorch, TensorFlow, Selenium, Flask, Pandas, NumPy, scikit-learn), Java, SQL, C/C++, HTML/CSS, JavaScript, LaTeX

Developer Tools/Technologies/Frameworks: VS Code, Eclipse, Linux, Git, GitHub, MariaDB, Jupyter Notebook

Certificates: Amazon Web Services Certified Solutions Architect Associate

Leadership / Extracurricular

Extracurriculars: Asian Intervarsity, WolvSec, Michigan Quizbowl, 3D Printing Club